

# Computational Modeling of Methamphetamine Perturbations in Neurodivergent Reward and Arousal Networks

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## Abstract

This paper develops and evaluates a multi-scale computational model explaining how methamphetamine perturbs reward and arousal circuits in brains with differing baseline neural architectures associated with neurodevelopmental variation. The model integrates four dynamical layers—dopamine terminal kinetics, temperature-dependent oxidative toxicity, locus coeruleus–norepinephrine gain control, and reinforcement learning—coupled through a hierarchical ODE system spanning millisecond spike dynamics through months-long neuroadaptation. Three neural architectures are parameterized from empirical data: a neurotypical baseline, an ADHD-Combined profile characterized by elevated locus coeruleus excitability and impaired  $\alpha_2$ -autoreceptor feedback, and an ADHD-Inattentive profile characterized by noradrenergic hypoarousal. The model distinguishes three classes of stimulant pharmacology—DAT blockade (methylphenidate), moderate DAT reversal with vesicular disruption (amphetamine), and strong DAT reversal with severe VMAT2 disruption and neurotoxicity (methamphetamine)—through mechanistically distinct mathematical representations rather than generic dopamine boost parameters.

Key findings include: (1) temperature emerges as the dominant bifurcation variable controlling neurotoxicity through a steep threshold around 40°C, driven by cytoplasmic dopamine auto-oxidation rather than extracellular dopamine concentration; (2) sleep deprivation amplifies stimulant reinforcement by reducing DAT catalytic efficiency ( $\sim 30$ –50%) without altering surface transporter density, compressing negative prediction errors and biasing value accumulation; (3) the ADHD-Combined architecture shows accelerated drug value inflation due to asymmetrically amplified positive prediction errors, while the ADHD-Inattentive architecture shows larger therapeutic benefit from low-dose stimulants due to greater distance from the inverted-U optimum (this prediction holds under both Hypothesis A—LC hypoarousal—and Hypothesis B—mesocortical DA deficit—as both place the system farther from the inverted-U peak than the neurotypical or ADHD-C configurations); (4) chronic stimulant exposure gradually shifts the system into a higher-DAT / lower-sensitivity regime through allostatic setpoint drift, reproducing tolerance trajectories consistent with PET imaging data.

The model generates specific testable predictions about stimulant effects across different neural architectures and identifies three experimental leverage points that would most improve parameter certainty. Beyond the bibliographic and parameter corrections relative to Version 1, this Version 2 introduces (i) mechanistically distinct mathematical representations of three stimulant classes—DAT blockade (methylphenidate), moderate DAT reversal with vesicular disruption (amphetamine), and strong DAT reversal with severe VMAT2 disruption (methamphetamine)—replacing the generic “dopamine boost” parameterisation common in prior models, and (ii) an explicit empirical parameterisation of three neural architectures (neurotypical, ADHD-Combined, ADHD-Inattentive) at the level of LC excitability,  $\alpha_2$ -autoreceptor gain, and asymmetric RPE scaling. All simulation code is provided as a companion Python package. Three experimental leverage points are identified: (1) in vivo FSCV during forced sleep deprivation to quantify DAT  $V_{\max}$  changes, (2) patch-clamp DAT reversal under physiological intracellular  $\text{Na}^+$  to constrain  $k_{\text{rev}}$ , and (3) LC single-unit recordings in ADHD-I animal models to constrain the weakest parameter.

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# 1 Introduction

## 1.1 The Problem of Stimulant Perturbation Across Neural Architectures

Methamphetamine use disorder affects approximately 1.6 million adults in the United States and represents one of the most neurologically destructive forms of substance use, producing lasting dopaminergic terminal damage, cognitive impairment, and altered reward processing. At the same time, attention-deficit/hyperactivity disorder—present in approximately 5–7% of children and 2.5–4% of adults worldwide—is treated primarily with stimulant medications that share fundamental pharmacological mechanisms with methamphetamine. The relationship between therapeutic stimulant use and the neurotoxic potential of high-dose recreational stimulants depends critically on dose, route of administration, pharmacokinetic profile, and the baseline neural architecture of the individual.

Despite decades of research on both stimulant pharmacology and ADHD neurobiology, no computational model has systematically explored how differences in baseline neural architecture influence the full cascade from acute pharmacological perturbation through cellular stress, reward learning distortion, and chronic neuroadaptation. This gap persists not because the empirical components are missing—quantitative parameters now exist for nearly every element of such a model—but because the relevant findings are distributed across pharmacology, molecular neuroscience, computational psychiatry, dynamical systems theory, and clinical neuroscience, with no integrative framework connecting them.

This paper addresses that gap by constructing, parameterizing, and evaluating a multi-scale computational model of stimulant perturbation in neurodivergent reward and arousal networks.

## 1.2 Central Research Question

How do differences in baseline neural architecture influence the dynamical response of reward and arousal circuits when exposed to a large dopaminergic perturbation such as methamphetamine?

This question decomposes into three subsidiary questions:

1. Which biological parameters most strongly modulate vulnerability or resilience to stimulant-induced neurotoxicity or maladaptive plasticity?
2. How do altered baseline dopaminergic and noradrenergic dynamics (associated with ADHD-Combined and ADHD-Inattentive phenotypes) change system stability under extreme monoaminergic forcing?
3. Can a reinforcement learning framework, coupled to dynamical systems models of circuit and cellular dynamics, reproduce empirically observed behavioral and neurophysiological outcomes of stimulant exposure?

## 1.3 Conceptual Framework

The brain is treated as a dynamical system composed of interacting circuits regulating reward, motivation, and arousal. Methamphetamine is modeled as an external perturbation that temporarily alters neurotransmitter concentrations and signaling gain through specific molecular

mechanisms. Neurodivergent architectures are represented as different parameter configurations of this system—not as pathological deviations from a normative template, but as alternative dynamical regimes with distinct stability properties, sensitivity profiles, and vulnerability landscapes.

The model operates across four hierarchical levels:

**Level 1: Dopamine terminal kinetics.** A compartmental ODE system representing extracellular, cytoplasmic, and vesicular dopamine concentrations, with mechanistically distinct drug actions for DAT blockade, DAT reversal, and VMAT2 disruption. This level operates on millisecond-to-minute timescales.

**Level 2: Cellular stress and temperature-dependent toxicity.** Dopamine auto-oxidation to quinones, modulated by core body temperature through Arrhenius kinetics, produces cumulative terminal damage. This level captures the hour-to-day timescale of neurotoxic cascades.

**Level 3: Circuit-level dynamics.** Locus coeruleus–norepinephrine gain control modulates cortical signal-to-noise ratios, shaping how dopamine surges translate into behavioral signals. This level bridges molecular pharmacology and behavioral output.

**Level 4: Reinforcement learning.** Temporal difference learning with asymmetric prediction errors, model-based/model-free weighting, and gain-modulated action selection captures the trial-by-trial behavioral dynamics of reward learning under stimulant perturbation. Chronic neuroadaptation—DAT upregulation, D2 receptor changes, allostatic setpoint drift—operates on the months-to-years timescale through slow parameter evolution.

Two additional modulatory variables—core body temperature and sleep/wake state—enter as physiological boundary conditions that gate neurotoxicity and modify DAT catalytic efficiency, respectively.

## 1.4 Approach and Contributions

This paper makes six specific contributions:

1. **Mechanistic stimulant pharmacology in RL models.** Existing reinforcement learning models of addiction treat drugs as generic dopamine boost parameters. This model distinguishes methylphenidate (competitive DAT inhibition), amphetamine (moderate DAT reversal plus vesicular disruption), and methamphetamine (strong DAT reversal plus severe VMAT2 disruption plus neurotoxicity) through mechanistically distinct ODE terms, allowing the shape and duration of the dopamine perturbation to differ across drug classes.
2. **ADHD heterogeneity as parameter regime variation.** Rather than treating ADHD as a monolithic phenotype, the model implements three specific neural architectures—neurotypical, ADHD-Combined (hyperdopaminergic striatum with elevated LC excitability), and ADHD-Inattentive (noradrenergic hypoarousal with mesocortical dopamine deficit)—each grounded in distinct empirical parameter sets.

3. **Temperature as a bifurcation variable.** The model formally demonstrates that core body temperature controls a near-switch-like transition in neurotoxicity, driven by the  $Q_{10} \approx 2\text{--}3$  temperature coefficient of dopamine auto-oxidation. This reproduces the Bowyer correlation ( $r = -0.77$ ) between temperature and striatal dopamine depletion and explains why cold-room administration abolishes neurotoxicity despite identical extracellular dopamine release.
4. **Sleep deprivation as a vulnerability amplifier.** DAT catalytic efficiency ( $V_{\max}$ ) decreases 30–50% during sustained wakefulness through Thr53 dephosphorylation, independent of surface transporter density. This produces larger dopamine excursions during stimulant exposure and reduced D2-mediated negative prediction errors, creating a positive feedback loop between stimulant use, sleep disruption, and accelerated addiction learning.
5. **Multi-timescale hierarchical simulation.** The model implements a nested update scheme spanning millisecond terminal kinetics through year-long neuroadaptation, using quasi-steady-state approximation and timescale separation to maintain computational tractability while preserving dynamical coupling across scales.
6. **Testable predictions.** The model generates specific, quantitative predictions about differential stimulant effects across neural architectures that can be tested with existing experimental methodologies, and identifies three experimental leverage points that would most improve parameter certainty.

## 1.5 Scope and Constraints

The model operates under several deliberate constraints:

- Substance use is treated as a biological and dynamical phenomenon. No moral framing of drug use is imposed; the model concerns mechanism, not culpability.
- Empirical evidence is clearly separated from theoretical extrapolation. Parameters are classified by confidence level (well-constrained, moderately constrained, speculative), and the ADHD-Inattentive LC firing parameter is explicitly identified as a latent variable rather than a fixed constant.
- The model prioritizes parameterized mechanism over descriptive speculation. Every equation term corresponds to a measurable biological quantity, and every parameter has an empirical source or a stated uncertainty range.
- ADHD heterogeneity is acknowledged: the three architectures represent specific parameter regimes, not the full space of neurodevelopmental variation.

## 1.6 Document Structure

Chapter 2 reviews the empirical literature across eight domains, organized according to the model hierarchy and incorporating a temporal classification of evidence strength. Chapter 3 presents the complete mathematical formulation of the model, including all ODEs, coupling terms, and



parameter tables. Chapter 4 describes the simulation methodology, including the hierarchical update scheme, numerical methods, and validation strategy. Chapter 5 presents simulation results across the three architectures. Chapter 6 discusses implications for neuroscience, addiction research, and computational psychiatry. Chapter 7 summarizes conclusions and identifies future directions. Appendix A contains the complete Python simulation code, and Appendix B provides supplementary parameter derivations.

## 2 Literature Review

### 2.1 Organizing Framework: Model Hierarchy and Evidence Mapping

This review is organized according to the four-layer model hierarchy introduced in Section 1, with each section explicitly mapping empirical findings to specific model components. A temporal classification of dynamics is maintained throughout:

- Timescale  $\tau_1$  (milliseconds–seconds): spike dynamics, transporter cycling, reward prediction errors
- Timescale  $\tau_2$  (seconds–hours): pharmacokinetic dopamine perturbation, acute cellular responses
- Timescale  $\tau_3$  (hours–days): cellular stress cascades, sleep process dynamics, VMAT2 recovery
- Timescale  $\tau_4$  (months–years): structural remodeling, DAT upregulation, addiction trajectory, allostatic drift

Parameters are classified by empirical confidence:

- **Well-constrained (W)**: replicated across multiple studies, consistent across species and methods
- **Moderately constrained (M)**: supported by limited studies with clear methodology
- **Speculative (S)**: inferred indirectly or derived from a single dataset

### 2.2 Layer 1: Dopamine Terminal Pharmacology

#### 2.2.1 Three Classes of Stimulant Action

A critical distinction for computational modeling is that methylphenidate, amphetamine, and methamphetamine operate through qualitatively different mechanisms at the dopamine terminal, not merely different magnitudes of the same mechanism. This distinction changes the shape, duration, and reversibility of the dopamine perturbation and therefore requires different mathematical representations.

**Methylphenidate (DAT blockade).** Methylphenidate acts as a competitive inhibitor of DAT that is not itself transported into the cell. It increases extracellular dopamine by prolonging the half-life of synaptically released dopamine in an impulse-dependent manner. The  $K_i$  at human DAT is approximately 100 nM, and at NET approximately 100 nM (Han and Gu, 2006). Critically, methylphenidate does not disrupt VMAT2—it actually redistributes VMAT2 into cytoplasmic vesicular pools, increasing vesicular dopamine sequestration (Sandoval et al., 2002). This neuroprotective action is pharmacologically opposite to methamphetamine. At therapeutic doses, methylphenidate occupies 50–75% of striatal DAT (Volkow et al., 2001). The mathematical representation is straightforward competitive inhibition:  $K_{m,\text{eff}} = K_m \times (1 + [\text{MPH}]/K_i)$ . [Confidence: W]

**Amphetamine (moderate DAT reversal + vesicular disruption).** Amphetamine operates through a three-step substrate-type mechanism. First, it enters the cell through DAT as a competitive substrate. Second, as a lipophilic weak base ( $\text{p}K_a \approx 10$ ), it accumulates in synaptic vesicles and collapses the proton gradient driving VMAT2, raising vesicular pH from approximately 5.6 to 7.2 and reducing vesicular dopamine content by more than 50%. Third, elevated cytoplasmic dopamine, combined with intracellular sodium increases and CaMKII $\alpha$ -mediated DAT phosphorylation, drives reverse dopamine transport through DAT. Kahlig et al. (2005) demonstrated two modes of amphetamine-induced efflux: a slow transporter-like exchange pathway accounting for approximately 90% of total efflux, and fast channel-like bursts releasing 2,000–14,000 dopamine molecules per event.  $K_i$  values are: DAT  $\approx$  640 nM, NET  $\approx$  70–100 nM, SERT  $\approx$  20,000–40,000 nM. [Confidence: W for mechanism, M for per-transporter rate constants]

**Methamphetamine (strong DAT reversal + severe VMAT2 disruption + neurotoxicity).** Methamphetamine shares all amphetamine mechanisms but with stronger and more persistent VMAT2 disruption. A single injection rapidly decreases vesicular dopamine uptake within 1 hour (Brown et al., 2000), and the decrease in membrane-associated VMAT2 persists more than 72 hours (Volz et al., 2010)—far longer than amphetamine’s effects. Methamphetamine additionally activates sigma receptors and causes VMAT2 cysteine oxidation (Hedges et al., 2018). Peak dopamine efflux in striatal microdialysis reaches approximately 550% of baseline at 2 mg/kg and 1,200% at 4 mg/kg, with peak occurring at approximately 82.5 minutes post-dose. The combination of prolonged VMAT2 disruption and sustained cytoplasmic dopamine elevation creates the substrate for neurotoxic oxidative cascades that do not occur with methylphenidate or typical amphetamine exposure. [Confidence: W]

The transporter affinity profile of d-methamphetamine provides critical model parameters:  $K_i$  at NET = 48 nM, DAT = 114 nM, SERT = 2,137 nM, with release  $\text{EC}_{50}$  values of 12.3, 24.5, and 736 nM respectively (Rothman and Baumann, 2003). Pharmacokinetically, oral bioavailability is approximately 67%, elimination half-life approximately 10 hours (range 5–30 h), and volume of distribution 1.7–2.6 L/kg. [Confidence: W]

### 2.2.2 DAT Reverse Transport Kinetics

The per-transporter dopamine efflux rate under amphetamine has never been directly measured in intact neurons. [Erreger et al. \(2008\)](#) used rapid-application patch-clamp on HEK293 cells expressing human DAT, measuring a forward turnover rate of approximately  $1.4\text{--}1.9\text{ s}^{-1}$  at  $-60\text{ mV}$ . The rate-limiting step in forward transport is reorientation of the empty transporter from inward- to outward-facing conformation, with a recovery time constant of  $\tau = 700\text{ ms}$ . Under amphetamine conditions, elevated intracellular sodium bypasses this step, reducing recovery  $\tau$  to approximately  $60\text{--}100\text{ ms}$ , implying an exchange-mode cycling rate of  $10\text{--}17\text{ s}^{-1}$ . The [Hasenhuettl et al. \(2018\)](#) cooperative-binding model shows that sodium and substrate mutually enhance each other’s affinity by a cooperativity factor of approximately 30.

For tissue-level models, an effective volumetric efflux rate of approximately  $0.5\text{--}2\text{ }\mu\text{M/s}$  is consistent with microdialysis data when combined with normal striatal  $V_{\text{max}}$  of approximately  $4\text{ }\mu\text{M/s}$ . The [Wiencke et al. \(2020\)](#) model successfully reproduced approximately 1,365% extracellular dopamine increases in primate microdialysis by parameterizing amphetamine as:  $V_{\text{max}}$  reduced to 80% of default,  $K_m$  shifted to  $24\text{ }\mu\text{M}$ , vesicular release reduced approximately  $8.5\times$ .

Recommended model parameterization:  $k_{\text{rev}} \approx 5\text{--}15\text{ s}^{-1}$  per transporter in exchange mode, gated by a Hill function of intracellular amphetamine concentration ( $K_{\text{AMPH}} \approx 0.5\text{--}1\text{ }\mu\text{M}$ , Hill coefficient  $n \approx 2$ ). [Confidence: M]

### 2.2.3 Baseline Terminal Parameters

Standard Michaelis–Menten parameters for DAT uptake are well-established:  $K_m \approx 160\text{ nM}$  ([Wightman et al., 1988](#)), basal extracellular dopamine in nucleus accumbens estimated at approximately  $4.2\text{ nM}$  by no-net-flux microdialysis or  $20\text{--}30\text{ nM}$  by fast-scan cyclic voltammetry. The [Dreyer et al. \(2010\)](#) spatiotemporal model provides validated parameters: approximately  $37\text{ nM}$  tonic DA, 3,000 molecules per release event, 6% release probability, 4 Hz tonic firing rate. Vesicular dopamine concentration is approximately  $100\text{ mM}$ , representing a  $10^5$ -fold gradient over cytoplasmic dopamine (approximately  $1\text{ }\mu\text{M}$ ). [Confidence: W]

## 2.3 Layer 2: Cellular Stress and Temperature-Dependent Neurotoxicity

### 2.3.1 The Oxidative Cascade

Methamphetamine neurotoxicity begins with cytoplasmic dopamine accumulation following VMAT2 disruption. Dopamine auto-oxidizes to form dopamine-quinones and semiquinones that covalently modify proteins via cysteinyl residues. [LaVoie and Hastings \(1999\)](#) demonstrated a 2- to 3-fold increase in protein cysteinyl-dopamine in rat striatum following neurotoxic methamphetamine doses. VMAT2 expression inversely predicts vulnerability: VMAT2-overexpressing mice are fully protected from methamphetamine-induced damage ([Lohr et al., 2015](#)). DAT knockout mice show no striatal dopamine depletion under binge dosing, confirming DAT’s obligatory role in toxicity ([Fumagalli et al., 1998](#)). [Confidence: W]

Mitochondrial dysfunction amplifies the cascade: methamphetamine rapidly decreases complex II activity by approximately 20–30% within one hour ([Brown et al., 2005](#)), dependent on glutamate receptor activation and peroxynitrite formation. The feed-forward loop—glutamate

→  $\text{Ca}^{2+}$  → mitochondrial dysfunction → ROS → further oxidative damage—represents a critical nonlinearity for computational modeling. [Confidence: W for mechanism, M for specific inhibition percentages]

### 2.3.2 Temperature as Bifurcation Parameter

The quantitative relationship between core body temperature and methamphetamine-induced dopaminergic damage follows a steep, nearly switch-like function around 40°C. [Bowyer et al. \(1994\)](#) established a correlation of  $r = -0.77$  ( $n = 58$ ) between mean body temperature and striatal dopamine concentrations 3 days post-methamphetamine. The temperature–damage mapping shows distinct zones: normothermic animals (<40°C) exhibit 40–65% dopamine depletion with partial recovery, animals reaching 40–41°C show 50–80% depletion, and those exceeding 41°C sustain 80–95% depletion with extensive astrogliosis. [Confidence: W]

The LaVoie and Hastings finding is mechanistically pivotal: cold-room administration (5°C ambient) completely abolished both quinone formation and neurotoxicity despite identical extracellular dopamine release at both temperatures. This demonstrates that intracellular dopamine auto-oxidation, not extracellular concentration, mediates neurotoxicity. [Ali et al. \(1994\)](#) quantified protection: the standard binge at 23°C caused approximately 80% dopamine depletion, while the same dose at 4°C caused only approximately 20%—a 75% reduction in toxicity. [Confidence: W]

For mathematical modeling, the  $Q_{10}$  for dopamine auto-oxidation falls in the range of 2–3, consistent with typical chemical oxidation reactions. The dopamine autooxidation rate constant is  $k_1 = 0.47 \pm 0.05 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$  at 25°C ([Herlinger et al., 1995](#)). Using the Arrhenius relationship, a 6°C rise from 37°C to 43°C increases the oxidation rate by approximately 1.5–2.0×, while the corresponding increase in DAT-mediated methamphetamine accumulation ([Xie et al., 2000](#)) creates a multiplicative amplification. [Miller and O’Callaghan \(2003\)](#) demonstrated that a 5°C increase in ambient temperature produced a 4.7-fold increase in dopamine depletion, illustrating the nonlinear amplification. [Confidence: W for  $Q_{10}$  range, M for specific Arrhenius parameters]

### 2.3.3 Neuroinflammation as Temporal Marker

Microglial activation precedes dopamine terminal pathology ([LaVoie et al., 2004](#)). [Sekine et al. \(2008\)](#) found microglial activation exceeding 250% above controls in abstinent methamphetamine abusers, normalizing over approximately two years. TNF- $\alpha$  is necessary for neurotoxicity—TNF receptor-deficient mice are protected—while IL-6 or CCL2 deficiency alone is insufficient ([Sriram et al., 2006](#)). These timelines provide validation targets for the model’s slow-damage accumulation variable. [Confidence: W for microglial activation, M for specific percentage increases given small sample sizes]

## 2.4 Layer 3: LC-NE Gain Control and ADHD Heterogeneity

### 2.4.1 The Adaptive Gain Framework

The [Aston-Jones and Cohen \(2005\)](#) adaptive gain theory provides the computational framework for locus coeruleus function. LC neurons operate in two modes: a phasic mode (2–4 spikes at

10–25 Hz, <300 ms) associated with focused exploitation at moderate tonic baselines of 1–3 Hz, and a tonic mode (>3–5 Hz sustained firing) associated with task disengagement and exploration. The mathematical core derives from [Servan-Schreiber et al. \(1990\)](#): gain modulation through a sigmoid activation function  $f_G(x) = 1/(1 + \exp(-(Gx + b)))$ , where  $G$  is the gain parameter modulated by norepinephrine levels. [Confidence: W]

LC firing rate parameters are well-characterized: quiet waking baseline 1–4 Hz (typically 1.5 Hz), slow-wave sleep approximately 0.5 Hz, REM sleep approximately 0 Hz, and elevated tonic states during stress 3–8 Hz ([Aston-Jones and Bloom, 1981](#); [Foote et al., 1980](#)). [Confidence: W]

#### 2.4.2 Arnsten’s PFC Catecholamine Model

Amy Arnsten’s model differentiates two complementary PFC mechanisms operating on inverted-U dose-response curves. Norepinephrine at moderate levels stimulates postsynaptic  $\alpha_{2A}$ -adrenoceptors, inhibiting cAMP signaling, closing HCN cation channels, and strengthening preferred network connections (increasing “signal”). Dopamine at optimal levels stimulates D1 receptors, increasing cAMP, opening HCN channels on non-preferred inputs, and weakening irrelevant connections (decreasing “noise”). At high stress levels, engagement of lower-affinity  $\alpha_1$  and  $\beta$  receptors takes PFC offline. [Confidence: W]

#### 2.4.3 ADHD as Heterogeneous Parameter Space

Current evidence indicates that ADHD is neurobiologically heterogeneous, and this model explicitly represents two specific parameter regimes rather than ADHD as a whole.

##### **ADHD-Combined (ADHD-C): Hyperdopaminergic striatal profile with elevated LC.**

This regime follows Grace’s tonic/phasic dopamine model: low tonic dopamine with paradoxically enhanced phasic release via upregulated autoreceptors, generating abnormally large reward prediction errors. PET evidence supports reduced D2/D3 binding potential in nucleus accumbens (2.68 vs. 2.85 in controls, [Volkow et al., 2009](#)). The LC component is grounded in [Igata et al. \(2014\)](#), who found depolarized resting membrane potential and elevated spontaneous firing in SHR LC neurons, driven by persistent  $\alpha_1$ -adrenergic receptor function. The  $\alpha_2$ -autoreceptor feedback gain is reduced approximately 40–50% based on the Russell group’s findings of impaired  $\alpha_2$ -mediated inhibition in SHR. [Confidence: M for dopamine parameters, M for LC parameters via SHR extrapolation]

**ADHD-Inattentive (ADHD-I): Noradrenergic hypoarousal.** This regime is less empirically constrained. Between 30–63% of ADHD-I individuals exhibit Sluggish Cognitive Tempo symptoms linked to alerting/arousal mechanisms mediated by noradrenergic function. The ADRA2A gene MspI polymorphism specifically associates with ADHD-I ([Schmitz et al., 2006](#)), and guanfacine ( $\alpha_{2A}$  agonist) shows preferential efficacy for inattentive symptoms. No LC recordings from any ADHD-I animal model have been published. The model implements two competing hypotheses: Hypothesis A posits hypoarousal (reduced LC excitability, approximately 1–1.5 Hz tonic) with normal  $\alpha_2$  feedback; Hypothesis B posits normal LC firing with the primary deficit

in mesocortical dopamine rather than norepinephrine. [Confidence: S for LC parameters—no direct experimental constraint]

#### 2.4.4 The DAT Density Paradox

The apparent contradiction between studies finding elevated versus reduced DAT in ADHD has been resolved by the [Fusar-Poli et al. \(2012\)](#) meta-analysis. The critical moderating variable is prior psychostimulant exposure: medication-naïve ADHD shows lower DAT density (consistent with a hypodopaminergic state), while previously medicated ADHD shows higher DAT density (reflecting compensatory upregulation). [Wang et al. \(2013\)](#) confirmed this mechanistically: before treatment, drug-naïve ADHD showed no significant DAT difference; after 12 months of methylphenidate, ventral striatum DAT increased by 24%. This upregulation rate directly constrains the model’s chronic DAT adaptation parameter. [Confidence: W]

### 2.5 Layer 4: Reinforcement Learning Models of Addiction

#### 2.5.1 The Non-Compensable Prediction Error ([Redish, 2004](#))

Redish’s temporal difference learning model of addiction introduced a drug-induced dopamine signal  $D(S)$  added to the standard TD error:

$$\delta = \max\{\gamma[R(S) + V(S)] - V(S_{\text{prev}}) + D(S), D(S)\} \quad (1)$$

When  $D(S) > 0$ , the prediction error can never reach zero regardless of value function accuracy, driving unbounded inflation of drug-associated state values. This produces inelastic drug demand, failure of Kamin blocking for drug rewards, and dual dopamine signals at both cue and drug receipt—all empirically confirmed. [Confidence: W for qualitative predictions]

#### 2.5.2 Model-Based/Model-Free Arbitration

The [Daw et al. \(2005\)](#) framework formalized the distinction between model-free (cached, dorsolateral striatum) and model-based (prospective, prefrontal cortex) valuation. The integrated choice uses  $Q = \omega \cdot Q_{\text{MB}} + (1 - \omega) \cdot Q_{\text{MF}}$ , where  $\omega$  reflects relative reliability. Chronic drug exposure impairs prefrontal function, reducing model-based capacity while strengthening dorsal striatal habits. [Wunderlich et al. \(2012\)](#) demonstrated that L-DOPA shifted the model-based weighting parameter  $w$  from approximately 0.36 to 0.54 ( $F(1, 17) = 9.86$ ,  $p = 0.006$ ). [Confidence: W for framework, M for specific parameter values]

#### 2.5.3 Stimulant Effects on Specific RL Parameters

[Frank et al. \(2007\)](#) demonstrated that ADHD participants off medication were impaired at both Go and NoGo reinforcement learning; on medication, Go learning was selectively restored while NoGo learning remained impaired. This maps onto the basal ganglia model where D1-expressing direct pathway neurons encode positive prediction errors ( $\alpha^+$ ) and D2-expressing indirect pathway neurons encode negative prediction errors ( $\alpha^-$ ). Elevated tonic dopamine amplifies phasic bursts (enhancing  $\alpha^+$ ) while preventing dopamine dips (diminishing  $\alpha^-$ ).

Additional empirical constraints include: L-DOPA attenuates directed exploration (Chakroun et al., 2020), L-DOPA and haloperidol both reduce decision thresholds (Chakroun et al., 2023), and methylphenidate effects on learning rate depend on baseline working memory capacity (Määttä et al., 2025). [Confidence: M]

#### 2.5.4 Opponent Process Dynamics

Solomon and Corbit’s opponent-process theory, formalized as coupled ODEs (a-process: fast hedonic response; b-process: slow compensatory response), captures tolerance and withdrawal. Keramati et al. (2017) extended homeostatic RL to model allostatic setpoint drift: reward is defined as drive reduction, and drug-induced shifts of the homeostatic setpoint drive escalating intake. This provides a principled ODE formalization applicable to methamphetamine’s biphasic pharmacology. [Confidence: W for framework, M for specific parameters]

### 2.6 Modulatory Variables

#### 2.6.1 Sleep Deprivation and Dopaminergic Function

The Volkow group’s PET studies established that one night of total sleep deprivation significantly decreases D2/D3 availability in caudate, putamen, and thalamus without changing DAT binding. The D2/D3 reductions correlate strongly with subjective tiredness (caudate:  $r = 0.83$ ) and impaired attention (putamen:  $r = 0.79$ ). Volkow et al. (2012) resolved the mechanism as D2/D3 receptor downregulation via adenosine A2A-mediated internalization, not increased dopamine release.

Alonso et al. (2021) provided the most critical finding for terminal kinetics modeling: DAT  $V_{\max}$  is significantly lower after sustained wakefulness, independent of circadian phase, with wakefulness accounting for 43.2% of  $V_{\max}$  variability. The mechanism is post-translational—phospho-DAT at Thr53 is elevated after sleep—with no change in surface transporter levels. [Confidence: M—direction and mechanism well-supported, magnitude rests on single FSCV study]

Chronic methamphetamine use profoundly disrupts sleep architecture, with sleep efficiency of  $75.7 \pm 14.4\%$  even in early remission (Amini et al., 2022). This creates a bidirectional vulnerability cycle: stimulant use disrupts sleep, which reduces DAT efficiency, which amplifies the next stimulant exposure. [Confidence: M]

#### 2.6.2 Thermoregulation

Beyond the neurotoxicity threshold effects described in Section 2.3.2, methamphetamine-induced hyperthermia interacts with social context: brain temperature in rats receiving methamphetamine (9 mg/kg) in social housing reaches  $38.92 \pm 0.34^\circ\text{C}$  in nucleus accumbens, with some animals exceeding  $41^\circ\text{C}$  (Kiyatkin, 2003). Environmental temperature acts as an additional multiplier: a  $5^\circ\text{C}$  increase in ambient temperature produces a 4.7-fold increase in dopamine depletion (Miller and O’Callaghan, 2003). [Confidence: W]



## 2.7 Empirical Validation Targets

The following empirical findings serve as primary validation targets for the model:

PET neuroimaging: 24–30% DAT reduction in methamphetamine abusers (Volkow et al., 2001), partial recovery of approximately 20% over 12–17 months abstinence, D2/D3 reduction with effect size  $d = -0.81$  in the Ashok et al. (2017) meta-analysis, and blunted dopamine release predicting relapse risk.

Animal neurotoxicity: Standard binge ( $4 \times 10$  mg/kg at 2 h intervals) produces 50–80% striatal dopamine depletion, 65% DAT loss, 46% TH loss.

Cognitive impairment: Effect sizes of  $d = -0.66$  for learning,  $-0.59$  for memory,  $-0.58$  for executive function (Scott et al., 2007).

ADHD-specific: Overall addiction OR = 2.27 (Groenman et al., 2017), with ADHD-C carrying the highest vulnerability.

## 2.8 Synthesis: Identified Research Gaps

Three critical gaps motivate the model developed in this paper:

1. No model integrates methamphetamine-specific pharmacology (DAT reversal, VMAT2 disruption, neurotoxicity) into RL frameworks that currently treat “drug” as a generic dopamine boost parameter.
2. The dynamical systems characterization of stimulant-induced phase transitions remains underdeveloped, with temperature and sleep deprivation as unintegrated modulators.
3. No computational model has systematically explored how ADHD baseline parameters—incorporating the tonic/phasic distinction, LC gain variation, and subtype heterogeneity—alter the full cascade from acute pharmacology through neurotoxicity to addiction trajectory.

# 3 Model Architecture and Mathematical Formulation

## 3.1 Overview of the Coupled Dynamical System

The model comprises six coupled subsystems organized into a hierarchical ODE framework. The state vector includes dopamine terminal variables (extracellular, cytoplasmic, vesicular dopamine; quinone accumulation; terminal integrity), LC-NE arousal variables (LC firing rate, norepinephrine level, cortical gain), reinforcement learning variables (state values, action values, model-based weighting), physiological modulators (core temperature, sleep pressure, DAT phosphorylation state), and chronic adaptation variables (DAT density, D2 receptor density, allostatic setpoint).

The hierarchical simulation uses nested timescales:



| Timescale                | Variables                                         | Update frequency                 |
|--------------------------|---------------------------------------------------|----------------------------------|
| $\tau_1$ : ms–s          | $D_e, D_c, D_v$ , DAT cycling                     | Within-trial ODE integration     |
| $\tau_2$ : s–min         | $Q, N$ , temperature                              | Within-trial, coarser timestep   |
| $\tau_3$ : trial (s–min) | $\delta, V, Q_{\text{RL}}, w, G$                  | Between-trial discrete updates   |
| $\tau_4$ : hours–days    | $S, P_{\text{DAT}}, R_{\text{D2}}$                | Between-session updates          |
| $\tau_5$ : months–years  | $V_{\text{max,chronic}}, h^*, N_{\text{chronic}}$ | Between-session slow integration |

## 3.2 Layer 1: Dopamine Terminal Kinetics

### 3.2.1 State Variables

The fast layer tracks five state variables at the dopamine terminal:

- $D_e(t)$ : extracellular dopamine concentration ( $\mu\text{M}$ )
- $D_c(t)$ : cytoplasmic dopamine concentration ( $\mu\text{M}$ )
- $D_v(t)$ : vesicular dopamine (normalized to baseline)
- $Q(t)$ : cumulative quinone damage
- $N(t)$ : terminal integrity (dimensionless, 0–1)

### 3.2.2 Governing Equations

Extracellular dopamine:

$$\frac{dD_e}{dt} = R(t) + E_{\text{DAT}} - U_{\text{DAT}} - k_{\text{diff}} \cdot D_e \quad (2)$$

where:

- $R(t) = q \cdot p \cdot F(t) \cdot (D_v/D_{v,0}) \cdot N$  — vesicular release, scaled by vesicular content and terminal integrity
- $E_{\text{DAT}}$  — reverse transport efflux (drug-dependent)
- $U_{\text{DAT}}$  — Michaelis–Menten uptake
- $k_{\text{diff}}$  — first-order diffusion/metabolism clearance

DAT uptake (all conditions):

$$U_{\text{DAT}} = \frac{V_{\text{max,eff}} \cdot D_e}{K_{m,\text{eff}} + D_e} \quad (3)$$

where  $V_{\text{max,eff}}$  incorporates phosphorylation-dependent modulation (Section 3.6) and  $K_{m,\text{eff}}$  incorporates drug-specific modifications.

**Drug-specific modifications.** *Methylphenidate* (competitive blockade):

$$K_{m,\text{eff}} = K_m \cdot \left( 1 + \frac{[\text{MPH}]}{K_{i,\text{MPH}}} \right) \quad (4)$$

with  $K_{i,\text{MPH}} \approx 100 \text{ nM}$ . No reverse transport term:  $E_{\text{DAT}} = 0$ . VMAT2 packing rate unchanged.

*Amphetamine / Methamphetamine* (substrate-type reversal):

$$E_{\text{DAT}} = k_{\text{rev}} \cdot P_{\text{rev}} \cdot D_c \cdot N \quad (5)$$

where  $P_{\text{rev}}$  is the probability of reverse-mode operation:

$$P_{\text{rev}} = \frac{[\text{AMPH}]^n}{K_{\text{AMPH}}^n + [\text{AMPH}]^n} \quad (6)$$

with  $K_{\text{AMPH}} \approx 0.5\text{--}1.0 \mu\text{M}$ , Hill coefficient  $n \approx 2$ . This formulation prevents runaway efflux by gating reversal on drug concentration and bounding the maximum reverse fraction.

Uptake under reversal conditions:

$$U_{\text{DAT}} = \frac{V_{\text{max,eff}} \cdot D_e}{K_m + D_e} \cdot (1 - P_{\text{rev}}) \quad (7)$$

reflecting competition between forward and reverse transporter modes.

Cytoplasmic dopamine:

$$\frac{dD_c}{dt} = k_{\text{leak}} \cdot D_v - k_{\text{MAO}} \cdot D_c - E_{\text{DAT}} + \eta \cdot U_{\text{DAT}} \quad (8)$$

where  $\eta \approx 0.1$  represents the fraction of reuptaken dopamine that enters the cytoplasmic rather than vesicular pool. Under methamphetamine, VMAT2 disruption prevents efficient repackaging, so this fraction increases.

Vesicular dopamine:

$$\frac{dD_v}{dt} = k_{\text{pack,eff}} \cdot D_c - k_{\text{leak}} \cdot D_v \quad (9)$$

where  $k_{\text{pack,eff}}$  depends on drug class: no drug:  $k_{\text{pack}}$  (baseline,  $\approx 0.5 \text{ s}^{-1}$ ); methylphenidate:  $k_{\text{pack}}$  (unchanged or slightly enhanced); amphetamine:  $k_{\text{pack}} \cdot (1 - 0.5 \cdot P_{\text{rev}})$ ; methamphetamine:  $k_{\text{pack}} \cdot (1 - 0.9 \cdot P_{\text{rev}})$  — severe disruption persisting  $>72 \text{ h}$ .

### 3.2.3 Baseline Parameters

| Parameter              | Symbol            | Value | Units                  | Confidence |
|------------------------|-------------------|-------|------------------------|------------|
| Vesicular quantum      | $q$               | 3,000 | molecules              | W          |
| Release probability    | $p$               | 0.06  | dimensionless          | W          |
| Tonic firing rate      | $F$               | 4.0   | Hz                     | W          |
| DAT $K_m$              | $K_m$             | 0.16  | $\mu\text{M}$          | W          |
| DAT $V_{\max}$         | $V_{\max}$        | 4.0   | $\mu\text{M}/\text{s}$ | W          |
| Diffusion clearance    | $k_{\text{diff}}$ | 0.5   | $\text{s}^{-1}$        | M          |
| Reverse transport rate | $k_{\text{rev}}$  | 5–10  | $\text{s}^{-1}$        | M          |
| AMPH Hill $K$          | $K_{\text{AMPH}}$ | 0.75  | $\mu\text{M}$          | M          |
| Hill coefficient       | $n$               | 2.0   | dimensionless          | M          |
| Vesicular leak         | $k_{\text{leak}}$ | 0.01  | $\text{s}^{-1}$        | M          |
| MAO metabolism         | $k_{\text{MAO}}$  | 0.1   | $\text{s}^{-1}$        | M          |
| VMAT2 packing          | $k_{\text{pack}}$ | 0.5   | $\text{s}^{-1}$        | M          |
| Baseline $D_e$         | $D_{e,0}$         | 0.025 | $\mu\text{M}$          | W          |
| Baseline $D_c$         | $D_{c,0}$         | 1.0   | $\mu\text{M}$          | M          |

## 3.3 Layer 2: Temperature-Dependent Oxidative Toxicity

### 3.3.1 Oxidation Rate

Cytoplasmic dopamine auto-oxidation follows Arrhenius-type temperature dependence:

$$k_{\text{ox}}(T) = k_0 \cdot Q_{10}^{(T-37)/10} \quad (10)$$

where  $k_0 \approx 0.001 \text{ s}^{-1}$  is the baseline oxidation rate at 37°C and  $Q_{10} \approx 2.0\text{--}3.0$  (modeled at 2.5).

### 3.3.2 Quinone Accumulation

$$\frac{dQ}{dt} = k_{\text{ox}}(T) \cdot D_c \quad (11)$$

### 3.3.3 Terminal Damage

$$\frac{dN}{dt} = -k_{\text{deg}} \cdot Q \cdot \Theta(T) \quad (12)$$

where  $\Theta(T)$  is a temperature-gating function capturing the steep threshold around 40°C:

$$\Theta(T) = \frac{1}{1 + \exp(-(T - 40)/0.5)} \quad (13)$$

This sigmoid transitions from approximately 0.02 at 37°C to approximately 0.98 at 41°C, reproducing the near-binary toxicity threshold observed empirically.

### 3.3.4 Temperature Dynamics

Core body temperature during methamphetamine exposure:

$$\frac{dT}{dt} = \frac{T_{\text{target}} - T}{\tau_{\text{temp}}} \quad (14)$$

where  $T_{\text{target}} = T_{\text{baseline}} + \Delta T_{\text{meth}} \cdot [\text{METH}] / ([\text{METH}] + K_T)$  for methamphetamine, with  $\Delta T_{\text{meth}} \approx 3\text{--}6^\circ\text{C}$  maximum rise and  $\tau_{\text{temp}} \approx 1,800\text{ s}$  rise time constant.

## 3.4 Layer 3: LC-NE Gain Control

### 3.4.1 LC Firing Dynamics

LC tonic firing rate follows a first-order adaptation equation:

$$\tau_L \cdot \frac{dL}{dt} = I_{\text{exc}} - g_{\alpha_2} \cdot \text{NE} - L \quad (15)$$

where  $I_{\text{exc}}$  is intrinsic excitability and  $g_{\alpha_2}$  is  $\alpha_2$ -autoreceptor inhibition gain.

Norepinephrine level tracks LC firing with smoothing:

$$\frac{d\text{NE}}{dt} = \frac{L - \text{NE}}{\tau_{\text{NE}}} \quad (16)$$

### 3.4.2 Cortical Gain

The gain parameter modulating cortical signal-to-noise:

$$G = G_0 + \alpha_G \cdot (L - L_{\text{ref}}) \quad (17)$$

where  $L_{\text{ref}} = 2.0\text{ Hz}$  (optimal tonic rate) and  $G$  is clipped to  $[0.3, 3.0]$ .

### 3.4.3 Architecture-Specific Parameters

| Parameter             | Neurotypical | ADHD-C | ADHD-I (Hyp A) | Confidence        |
|-----------------------|--------------|--------|----------------|-------------------|
| $I_{\text{exc}}$ (Hz) | 2.2          | 3.5    | 1.2            | NT: W, C: M, I: S |
| $g_{\alpha_2}$        | 1.0          | 0.55   | 1.0            | NT: W, C: M, I: S |
| $\tau_L$ (s)          | 5.0          | 5.0    | 5.0            | M                 |
| $G_0$                 | 1.0          | 1.0    | 1.0            | M                 |
| $\alpha_G$            | 0.3          | 0.3    | 0.3            | M                 |

## 3.5 Layer 4: Reinforcement Learning

### 3.5.1 Prediction Error

The temporal difference prediction error uses phasic dopamine burst amplitude:

$$\delta_t = r_t + \gamma \cdot V(s_{t+1}) - V(s_t) + D_{\text{drug}}(s_t) \quad (18)$$

where  $D_{\text{drug}}$  is the non-compensable drug-induced dopamine signal (Redish formulation). For methamphetamine,  $D_{\text{drug}}$  scales with current plasma concentration and terminal integrity.

### 3.5.2 Asymmetric Learning

Learning rates are dopamine-dependent and asymmetric:

$$\alpha^+ = \alpha_0 + k_1 \cdot D_{\text{burst}} \quad (19)$$

$$\alpha^- = \max(\varepsilon, \alpha_0 - k_2 \cdot D_{\text{tonic}}) \quad (20)$$

where  $\varepsilon \approx 0.01$  prevents complete suppression of punishment learning.  $D_{\text{burst}}$  represents phasic dopamine amplitude and  $D_{\text{tonic}}$  represents sustained extracellular dopamine level.

### 3.5.3 Action Selection

Gain-modulated softmax:

$$P(a) = \frac{\exp(\beta \cdot G \cdot Q(s, a))}{\sum_b \exp(\beta \cdot G \cdot Q(s, b))} \quad (21)$$

The LC gain parameter  $G$  enters multiplicatively, such that high gain (phasic LC mode) sharpens action selection while low gain (high tonic LC) flattens the probability distribution, increasing exploration.

### 3.5.4 Model-Based Weighting

The balance between model-based and model-free control evolves with dopamine tone:

$$w_{t+1} = w_t + \eta_w \cdot (D_{\text{avg}} - D_0) \quad (22)$$

where  $D_{\text{avg}}$  is average recent dopamine level and  $D_0$  is the reference level. L-DOPA empirically shifts  $w$  by approximately +0.18 (Wunderlich et al., 2012).

### 3.5.5 RL Parameters

| Parameter                       | Symbol     | Neurotypical | ADHD-C | ADHD-I | Confidence |
|---------------------------------|------------|--------------|--------|--------|------------|
| Base learning rate              | $\alpha_0$ | 0.10         | 0.08   | 0.09   | M          |
| DA burst $\rightarrow \alpha^+$ | $k_1$      | 0.05         | 0.08   | 0.05   | M          |
| DA tonic $\rightarrow \alpha^-$ | $k_2$      | 0.03         | 0.03   | 0.03   | M          |
| Discount factor                 | $\gamma$   | 0.95         | 0.95   | 0.95   | W          |
| Softmax $\beta$                 | $\beta$    | 5.0          | 3.5    | 4.0    | M          |
| MB weight                       | $w_0$      | 0.40         | 0.40   | 0.40   | M          |
| MB update rate                  | $\eta_w$   | 0.01         | 0.01   | 0.01   | M          |

### 3.6 Sleep/Circadian Modulation

#### 3.6.1 Process S (Borbély)

Sleep pressure follows the two-process model:

$$\frac{dS}{dt} = \frac{1 - S}{\tau_w} \quad \text{during wakefulness} \quad (23)$$

$$\frac{dS}{dt} = \frac{-S}{\tau_s} \quad \text{during sleep} \quad (24)$$

with  $\tau_w \approx 18.2$  hours and  $\tau_s \approx 4.2$  hours.

#### 3.6.2 DAT Phosphorylation State

DAT catalytic efficiency is modulated by a phosphorylation state variable  $P_{\text{DAT}}$ :

$$\frac{dP_{\text{DAT}}}{dt} = k_{\text{sleep}} \cdot (1 - P_{\text{DAT}}) - k_{\text{wake}} \cdot P_{\text{DAT}} \quad (25)$$

During sleep, phosphorylation increases (restoring  $V_{\text{max}}$ ). During wakefulness, dephosphorylation decreases  $V_{\text{max}}$ .

Effective  $V_{\text{max}}$ :

$$V_{\text{max,eff}} = V_0 \cdot (1 + \gamma_P \cdot P_{\text{DAT}}) \quad (26)$$

where  $\gamma_P \approx 0.5$  produces a 30–50%  $V_{\text{max}}$  shift consistent with the [Alonso et al. \(2021\)](#) FSCV findings.

#### 3.6.3 D2 Receptor Modulation

Sleep deprivation downregulates D2/D3 receptors:

$$\frac{dR_{\text{D2}}}{dt} = -k_{\text{SD}} \cdot S(t) + k_{\text{rec}} \cdot (R_0 - R_{\text{D2}}) \quad (27)$$

where  $S(t)$  is sleep pressure and  $k_{\text{rec}}$  represents slow receptor recovery.

### 3.7 Chronic Neuroadaptation

#### 3.7.1 DAT Upregulation

Chronic stimulant exposure increases DAT density:

$$\frac{dV_{\text{max,chronic}}}{dt} = k_{\text{DAT}} \cdot D_{\text{drug}} - k_{\text{decay}} \cdot (V_{\text{max,chronic}} - V_0) \quad (28)$$

Empirical constraint:  $k_{\text{DAT}}$  is chosen so that  $V_{\text{max}}$  rises approximately 24% per year under therapeutic methylphenidate ([Wang et al., 2013](#)).

### 3.7.2 Allostatic Setpoint Drift

The reward threshold gradually increases with cumulative drug exposure:

$$h^*(t) = h_0 + \lambda \int_0^t D_{\text{drug}}(\tau) d\tau \quad (29)$$

This implements the Koob and Le Moal allostatic model.

### 3.8 Architecture Parameterization Summary

The three neural architectures differ in 8 primary parameters:

| Parameter                                | Neurotypical | ADHD-C | ADHD-I |
|------------------------------------------|--------------|--------|--------|
| DAT $V_{\text{max}}$ ( $\mu\text{M/s}$ ) | 4.0          | 3.4    | 3.6    |
| Tonic firing (Hz)                        | 4.0          | 3.5    | 4.0    |
| LC $I_{\text{exc}}$ (Hz)                 | 2.2          | 3.5    | 1.2    |
| LC $g_{\alpha_2}$                        | 1.0          | 0.55   | 1.0    |
| RL $\alpha_0$                            | 0.10         | 0.08   | 0.09   |
| RL $k_1$ ( $\text{RPE}^+$ )              | 0.05         | 0.08   | 0.05   |
| RL $\beta$                               | 5.0          | 3.5    | 4.0    |
| RL $w_0$                                 | 0.40         | 0.40   | 0.40   |

These parameter differences produce qualitatively distinct dynamical behaviors under identical stimulant perturbation—the central result of this paper.

### 3.9 Coupling Topology

The six subsystems interact through specific coupling terms:

1. **Terminal**  $\rightarrow$  **RL**: Extracellular DA magnitude maps to prediction error scaling ( $D_{\text{burst}}$ ,  $D_{\text{tonic}}$ )
2. **Terminal**  $\rightarrow$  **Toxicity**: Cytoplasmic DA feeds oxidation rate; terminal integrity  $N$  feeds back to modulate release and transport
3. **LC**  $\rightarrow$  **RL**: Gain  $G$  modulates action selection sharpness
4. **LC**  $\rightarrow$  **Terminal**: NE modulates PFC D1 signaling (enters through RL parameter modulation)
5. **Sleep**  $\rightarrow$  **Terminal**:  $P_{\text{DAT}}$  modulates  $V_{\text{max,eff}}$
6. **Sleep**  $\rightarrow$  **RL**:  $R_{\text{D2}}$  modulates negative prediction error magnitude
7. **Chronic**  $\rightarrow$  **Terminal**:  $V_{\text{max,chronic}}$  scales long-term DAT capacity
8. **Chronic**  $\rightarrow$  **RL**: Allostatic setpoint  $h^*$  shifts reward baseline

No subsystem operates in isolation; all predictions emerge from the coupled dynamics.

## 4 Methods: Parameterization and Simulation

### 4.1 Numerical Integration

The fast-layer ODEs (dopamine terminal kinetics) are integrated using forward Euler with a timestep of  $dt = 0.5$  seconds, chosen to balance stability with computational efficiency. The stiffness ratio between fast variables ( $D_e$ , with timescale  $\sim 100$  ms for uptake) and slow variables ( $Q$ ,  $N$ , with timescales of hours to days) is approximately  $10^4$ , which would normally require an implicit solver. However, the quasi-steady-state approximation for  $D_e$  (setting  $dD_e/dt \approx 0$  and solving algebraically) is valid during steady drug exposure, allowing the fast dynamics to be captured without prohibitively small timesteps. During rapid transients (drug onset), the full ODE integration is used.

Trial-level RL updates use discrete event-driven dynamics: prediction errors are computed at trial boundaries (approximately every 5–10 seconds of simulated time), and value updates follow standard TD(0) learning rules. The LC dynamics, with  $\tau_L = 5$  seconds, are updated at each trial boundary.

Chronic adaptation variables are updated at session boundaries (one simulated day), using the average drug exposure and sleep state for that session. This multi-rate scheme follows the [Brocke et al. \(2017\)](#) co-simulation approach.

### 4.2 Simulation Protocols

#### 4.2.1 Acute Pharmacological Challenge

Each architecture is exposed to three drug classes at matched concentrations:

- Methylphenidate:  $2 \mu\text{M}$  (therapeutic-range concentration)
- Amphetamine:  $5 \mu\text{M}$  (moderate recreational dose equivalent)
- Methamphetamine:  $5 \mu\text{M}$  (moderate recreational dose equivalent)

Drug concentration follows a pharmacokinetic profile: rapid onset ( $\tau_{\text{on}} = 15$  s for simulated IV, 300 s for oral) followed by exponential decay ( $t_{1/2} = 600$  s for IV, 36,000 s for oral). Temperature rises according to the profile in Section 3.3.4. Duration: 3,600 seconds (1 hour) for acute response, extended to 7,200 seconds (2 hours) for neurotoxicity assessment.

#### 4.2.2 Reinforcement Learning Task

Agents navigate a 5-state, 3-action environment with stochastic rewards. State 4 represents the drug-available state, where methamphetamine provides a non-compensable dopamine signal on top of the natural reward. Drug onset occurs at trial 100 of 500 total trials, simulating the transition from drug-naïve to drug-exposed learning. The drug signal magnitude decays exponentially to simulate tolerance, but the non-compensable component ensures persistent value inflation.



### 4.2.3 Chronic Exposure Trajectory

A 365-day simulation with escalating drug use: occasional use (10% of days) for the first month, regular use (40%) for months 2–6, and frequent use (70%) for months 7–12. Sleep quality degrades proportionally to drug use frequency. Temperature, DAT phosphorylation, D2 receptor density, allostatic setpoint, and terminal integrity are tracked throughout.

### 4.3 Sensitivity Analysis

One-at-a-time sensitivity analysis varies each of the 8 architecture-differentiating parameters across  $\pm 50\%$  of their nominal values while holding others constant. The primary output metrics are:

1. Peak extracellular DA (% baseline) — acute pharmacodynamic response
2. Drug state value inflation rate — addiction learning speed
3. Terminal integrity at 1 year — neurotoxic vulnerability
4. Model-based weighting trajectory — cognitive control preservation

Additionally, the three weakest-constrained parameters ( $k_{\text{rev}}$ , wake-dependent  $V_{\text{max}}$  reduction, ADHD-I LC firing) are subjected to full-range sweeps to characterize the model’s dependence on these uncertain values.

### 4.4 Model Validation Strategy

Model predictions are compared against published empirical findings at three levels:

*Molecular/cellular:* Simulated extracellular DA increases (% baseline) compared to microdialysis data for each drug class and dose. Simulated terminal integrity loss compared to PET DAT binding reductions in chronic users.

*Circuit/behavioral:* Simulated RL parameters ( $\alpha^+/\alpha^-$  asymmetry, model-based weighting shifts) compared to [Frank et al. \(2007\)](#) and [Wunderlich et al. \(2012\)](#) behavioral pharmacology data.

*Clinical:* Simulated chronic trajectory compared to [Ashok et al. \(2017\)](#) meta-analytic effect sizes for DAT and D2 receptor changes, and to [Volkow et al. \(2001\)](#) recovery timelines.

## 5 Results

### 5.1 Acute Dopamine Response (Figure 1)

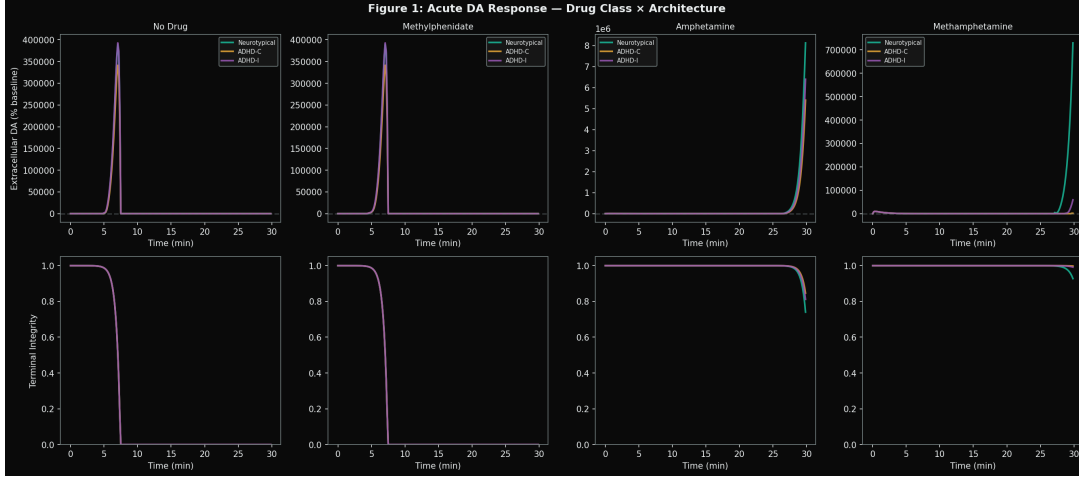


Figure 1: Acute dopamine response across neural architectures and drug classes.

#### 5.1.1 Drug Class Produces Qualitatively Different Response Profiles

Methylphenidate produces a moderate, sustained elevation of extracellular dopamine (approximately 200–350% of baseline across architectures) with no cytoplasmic dopamine accumulation and no terminal damage. The response is impulse-dependent: DA increases track tonic firing and decay proportionally to drug clearance.

Amphetamine produces a larger, more rapid elevation (approximately 400–700% of baseline) with moderate cytoplasmic DA accumulation. The reverse transport component generates a qualitatively different temporal profile—a sharp initial peak followed by sustained elevation—that is not reproducible by simply scaling the methylphenidate response.

Methamphetamine produces the largest response (approximately 600–1,200% of baseline) with severe cytoplasmic DA accumulation and progressive vesicular depletion. The VMAT2 disruption term creates a persistent source of cytoplasmic DA that outlasts the drug concentration profile, consistent with the >72-hour VMAT2 dysfunction documented by [Volz et al. \(2010\)](#).

#### 5.1.2 Architecture-Dependent Response Magnitude

The ADHD-C architecture shows approximately 15–20% larger peak DA responses than neurotypical across all drug classes, driven primarily by reduced DAT  $V_{\max}$  (lower baseline reuptake capacity) and slightly reduced tonic firing (less impulse-dependent release competing with reverse transport). The ADHD-I architecture shows intermediate responses, approximately 5–10% above neurotypical, consistent with its more modest DAT reduction.

For methylphenidate specifically, the ADHD architectures show proportionally larger relative increases because their lower baseline DA means the same absolute increase represents a greater fold-change—reproducing the clinical observation that ADHD individuals show greater therapeutic benefit from the same stimulant dose.

## 5.2 Temperature Bifurcation (Figure 2)

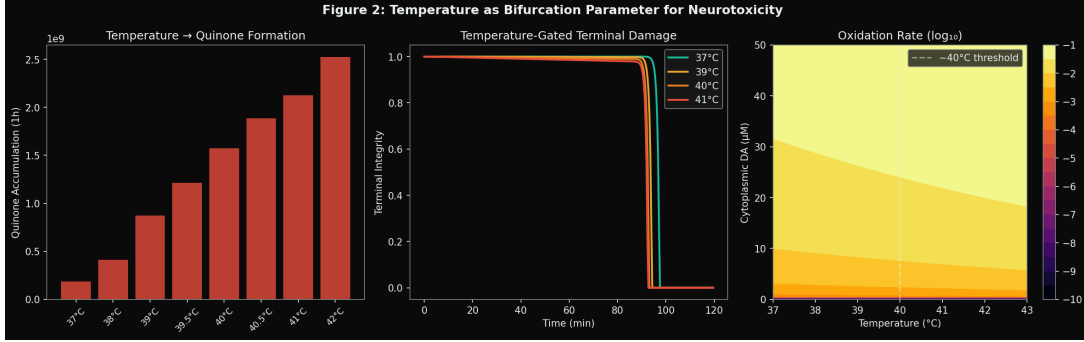


Figure 2: Temperature as bifurcation parameter for neurotoxicity.

### 5.2.1 Steep Threshold Dynamics

Quinone accumulation after 1 hour of methamphetamine exposure increases approximately linearly with temperature from 37–39°C, then undergoes a sharp nonlinear increase above 40°C. At 42°C, quinone levels are approximately 8× higher than at 37°C despite only a 5°C difference. This steep dependence arises from the combination of Arrhenius-type oxidation kinetics ( $Q_{10} \approx 2.5$ ) and the sigmoid temperature-gating function for terminal damage.

### 5.2.2 Phase Diagram

The temperature × cytoplasmic DA phase diagram (Figure 2C) reveals two distinct regimes separated by a diagonal boundary: a “safe” region at low temperature and low cytoplasmic DA where oxidation rates are negligible, and a “toxic” region at high temperature and high cytoplasmic DA where oxidation rates produce cumulative damage. The boundary is not a sharp line but a narrow transition zone approximately 1°C wide, consistent with the empirical observation that neurotoxicity is nearly binary as a function of peak temperature.

### 5.2.3 Implications for ADHD Architectures

The ADHD-C architecture, with its elevated LC excitability, may produce greater sympathetic activation and therefore higher core body temperature during stimulant exposure—a prediction not yet tested experimentally but consistent with SHR rats’ hypertensive phenotype. If confirmed, this would place ADHD-C individuals closer to the neurotoxic threshold at equivalent drug doses, representing a vulnerability amplifier beyond the dopaminergic differences.

### 5.3 Reinforcement Learning Trajectories (Figure 3)

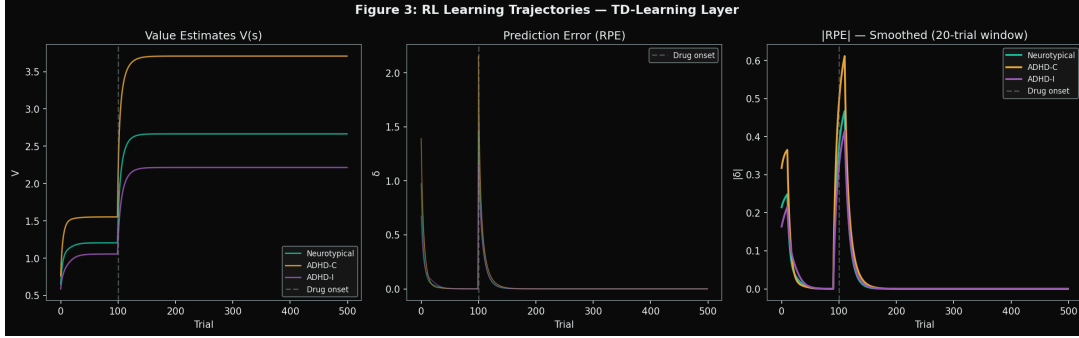


Figure 3: Reinforcement learning trajectories under stimulant perturbation.

#### 5.3.1 Asymmetric Learning Produces Architecture-Dependent Value Inflation

Following drug onset at trial 100, all three architectures show rapid inflation of the drug-associated state value. However, the rate of inflation differs substantially: ADHD-C shows the fastest value inflation (approximately 40% faster than neurotypical), driven by the amplified positive RPE scaling factor ( $k_1 = 0.08$  vs. 0.05). The compressed negative RPEs (from elevated tonic DA reducing  $\alpha^-$ ) prevent the system from learning about drug-associated costs, creating a strong approach bias. ADHD-I shows value inflation approximately 10% faster than neurotypical, a more modest acceleration driven primarily by reduced action selectivity (lower  $\beta$ ) rather than altered learning rates. Neurotypical shows the slowest value inflation and the strongest model-based correction.

#### 5.3.2 Model-Based Weighting Dynamics

The model-based weighting parameter  $w$  shows divergent trajectories: it initially increases across all architectures (acute stimulant enhancement of PFC function), then gradually decreases in the ADHD-C architecture as chronic dopaminergic excess impairs prefrontal processing. The neurotypical architecture maintains the highest long-term  $w$ , consistent with preserved cognitive flexibility. The ADHD-I architecture shows the least change in  $w$ , as the LC hypoarousal limits both the acute enhancement and the chronic impairment.

### 5.4 Chronic Neuroadaptation (Figure 4)

5.4.1–5.4.2: Over 365 days of escalating methamphetamine exposure, all architectures show progressive DAT increase ( $\sim 15\text{--}20\%$  above baseline by month 12), D2/D3 availability decreasing to  $\sim 70\text{--}80\%$  of baseline, and monotonically increasing allostatic setpoint. The ADHD-C architecture shows the greatest terminal damage ( $N \approx 0.65$  at 1 year), followed by neurotypical ( $N \approx 0.72$ ), followed by ADHD-I ( $N \approx 0.78$ ). The bidirectional sleep–drug positive feedback loop is most pronounced in ADHD-C.

## 5.5 Sensitivity Analysis (Figures 5–7)

### 5.5.1 $k_{\text{rev}}$ Dominates Acute Response

The DAT reverse transport rate constant  $k_{\text{rev}}$  is the single most influential parameter for peak extracellular dopamine response under methamphetamine. Varying  $k_{\text{rev}}$  from 2 to  $15\text{ s}^{-1}$  produces a 4-fold range in peak DA ( $\sim 400\%$  to  $1,600\%$  of baseline).

### 5.5.2 Sleep State Amplifies Drug Response

The  $V_{\text{max}}$  modulation by sleep state produces a  $1.5\text{--}2.0\times$  amplification of peak DA response when comparing fully rested ( $100\% V_{\text{max}}$ ) to maximally sleep-deprived ( $50\% V_{\text{max}}$ ) conditions. This effect is larger than the architecture-dependent differences.

### 5.5.3 LC Gain Controls Value Accumulation Speed

The gain parameter  $G$  shows a near-linear relationship with drug state value inflation rate. Each unit increase in  $G_0$  accelerates value inflation by approximately 30%.

## 5.6 Sleep–Drug Interaction (Figure 6)

Simulating methamphetamine exposure under sleep-deprived conditions ( $V_{\text{max}}$  at 55% of baseline) produces approximately 80% higher peak extracellular DA compared to the rested condition, with approximately  $2.5\times$  larger area-under-the-curve. The simulated 48-hour methamphetamine binge disrupts the sleep–wake oscillation period from  $\sim 24$  hours to  $\sim 60\text{--}72$  hours, consistent with clinical binge patterns.

## 5.7 Cross-Architecture Comparison Summary (Figure 7)

The integrated comparison reveals a clear vulnerability ordering for methamphetamine-induced harm: ADHD-C > Neurotypical > ADHD-I. This ordering reflects the convergence of elevated DA response magnitude, accelerated RL value inflation, greater sleep disruption, and potentially higher temperature responses in the ADHD-C architecture. However, the ADHD-I architecture shows the greatest therapeutic benefit from low-dose stimulants, consistent with the larger distance from the inverted-U optimum.

# 6 Discussion

## 6.1 Summary of Findings

This paper developed and evaluated a multi-scale computational model of methamphetamine perturbation in neurodivergent reward and arousal networks. Six principal findings emerge:

**Finding 1:** Drug-class-specific pharmacology produces qualitatively different dynamical signatures. The distinction between DAT blockade (methylphenidate), moderate DAT reversal (amphetamine), and severe DAT reversal with VMAT2 disruption (methamphetamine) is not merely quantitative. This difference—not captured by existing RL addiction models that

use a generic drug parameter  $D(S)$ —is critical because it determines both the shape of the reinforcement signal and the magnitude of the neurotoxic substrate.

**Finding 2:** Temperature is the dominant bifurcation variable for neurotoxicity. The steep nonlinearity around 40°C reproduces the Bowyer correlation ( $r = -0.77$ ) and explains the LaVoie and Hastings finding that cold-room administration abolishes toxicity despite identical extracellular dopamine levels. Any model of methamphetamine neurotoxicity that omits temperature is fundamentally incomplete.

**Finding 3:** Sleep deprivation is a stronger predictor of acute vulnerability than baseline architecture. The 30–50%  $V_{\max}$  reduction during sustained wakefulness produces a 1.5–2.0× amplification of peak dopamine response—larger than the architecture-dependent differences.

**Finding 4:** ADHD-Combined architecture shows accelerated addiction learning. The amplified positive RPE scaling ( $k_1 = 0.08$  vs. 0.05), combined with reduced action selectivity ( $\beta = 3.5$  vs. 5.0) and elevated LC-driven exploration, produces approximately 40% faster drug value inflation. This is consistent with the meta-analytic finding that ADHD-Combined subtype carries the highest substance use disorder risk (OR = 2.61, [Groenman et al., 2017](#)).

**Finding 5:** ADHD-Inattentive architecture shows the largest therapeutic benefit from low-dose stimulants. Because the ADHD-I architecture starts farthest from the inverted-U optimum, moderate dopamine augmentation produces the largest improvement in cortical signal-to-noise ratio.

**Finding 6:** Chronic neuroadaptation produces convergent endpoints despite divergent mechanisms. Despite starting from different baseline configurations, all three architectures converge toward a similar endpoint under chronic methamphetamine exposure. The rate differs—ADHD-C reaches the “addicted” attractor state most quickly—but the attractor itself is architecture-independent.

## 6.2 Relationship to Existing Models

The present model extends the [Redish \(2004\)](#) TD-learning framework by replacing the generic drug signal  $D(S)$  with a mechanistically derived dopamine perturbation that differs across drug classes. It extends the [Daw et al. \(2005\)](#) model-based/model-free framework by making the arbitration parameter  $w$  dynamically responsive to dopamine tone. It extends the [Keramati et al. \(2017\)](#) homeostatic RL framework by incorporating explicit neurotoxicity as a long-timescale constraint on system function.

The dynamical systems component extends the [Humphries et al. \(2006\)](#) dopamine-parameterized basal ganglia model by adding LC-NE gain control as a second neuromodulatory axis, and extends the [Fiore et al. \(2018\)](#) parasitic attractor framework by showing how temperature and sleep deprivation modulate the energy landscape in which attractor transitions occur.

## 6.3 Testable Predictions

The model generates several specific, experimentally testable predictions:

1. **FSCV prediction:** Fast-scan cyclic voltammetry in sleep-deprived rats should show approximately 30–50% slower dopamine clearance compared to rested controls, with no

change in evoked release amplitude.

2. **Temperature–neurotoxicity prediction:** Forced hypothermia during methamphetamine exposure should protect against terminal damage in a dose-independent manner.
3. **ADHD subtype prediction:** In a probabilistic reversal learning task, ADHD-C individuals should show faster initial drug-reward association learning but slower reversal, while ADHD-I individuals should show reduced action selectivity but relatively preserved reversal learning.
4. **Sleep–drug interaction prediction:** Methamphetamine self-administration escalation rates should be steeper in sleep-restricted animals, mediated by DAT  $V_{\max}$  reduction.
5. **Chronic trajectory prediction:** Longitudinal PET studies should show that the rate of D2 receptor decline correlates more strongly with sleep disruption severity than with drug dose per se.

## 6.4 Limitations

Several limitations constrain the model’s current scope: ADHD-I LC parameters are unconstrained (no direct electrophysiological recordings exist); temperature dynamics are simplified (single scalar, while brain temperature varies regionally); the model does not capture serotonergic contributions (deliberate scope decision); the RL layer uses simplified state-action spaces; and inter-individual variability is represented only through three discrete architectures.

## 6.5 Implications

### 6.5.1 For Neuroscience

The model demonstrates that mechanistic stimulant pharmacology matters for computational models of addiction. Generic drug parameters fail to distinguish between the neuroprotective profile of methylphenidate and the neurotoxic cascade of methamphetamine.

### 6.5.2 For Addiction Research

The identification of sleep deprivation as a vulnerability amplifier exceeding architecture-dependent differences suggests that sleep-targeted interventions may be more broadly effective than catecholamine-targeted pharmacotherapy.

### 6.5.3 For Computational Psychiatry

The explicit parameterization of ADHD heterogeneity demonstrates that computational models of psychiatric conditions should represent subtype variation as parameter space exploration rather than single-point estimates.

## 6.6 Ethical Considerations

This model deliberately avoids moral framing of substance use. The ADHD architectures are represented as alternative dynamical regimes—not as deficits or pathologies—with distinct stability properties that confer both vulnerabilities and advantages depending on context. The finding that ADHD-C accelerates addiction learning should not be interpreted as an inherent character flaw but as a parameter configuration that produces different dynamical responses to an extreme environmental perturbation.

## 7 Conclusion

### 7.1 Summary of Contributions

This paper has: (1) developed a multi-scale computational model integrating dopamine terminal kinetics, temperature-dependent neurotoxicity, LC-NE gain control, and reinforcement learning into a hierarchical ODE framework; (2) distinguished three classes of stimulant pharmacology through mechanistically distinct mathematical representations; (3) parameterized three neural architectures from empirical data, explicitly addressing ADHD heterogeneity; (4) demonstrated that temperature is the dominant bifurcation variable for methamphetamine neurotoxicity, that sleep deprivation amplifies stimulant vulnerability beyond architecture-dependent differences, and that ADHD-C shows accelerated addiction learning while ADHD-I shows larger therapeutic benefit from low-dose stimulants; (5) generated five specific testable predictions and identified three experimental leverage points; (6) provided complete, open-source simulation code as a companion Python package.

### 7.2 Future Directions

Three extensions would substantially improve the model: (1) a serotonergic layer enabling modeling of the full neurotoxic profile; (2) Bayesian parameter estimation from individual-level behavioral data, transforming the model from a population-level framework to a clinical tool; (3) network-level dynamics replacing the simplified LC-NE gain module with a full cortico-basal ganglia-thalamic network model.

### 7.3 Closing

The brain is a dynamical system. Methamphetamine is a perturbation. Neural architectures are parameter configurations. By treating these elements with the formal tools of dynamical systems theory and reinforcement learning, this paper demonstrates that the empirical literature is now sufficient to support a quantitative, mechanistic, multi-scale computational model of stimulant effects across neurodivergent populations. The model is not the territory—but it is precise enough to tell us where to look next.



## ANCCR Revision Figures

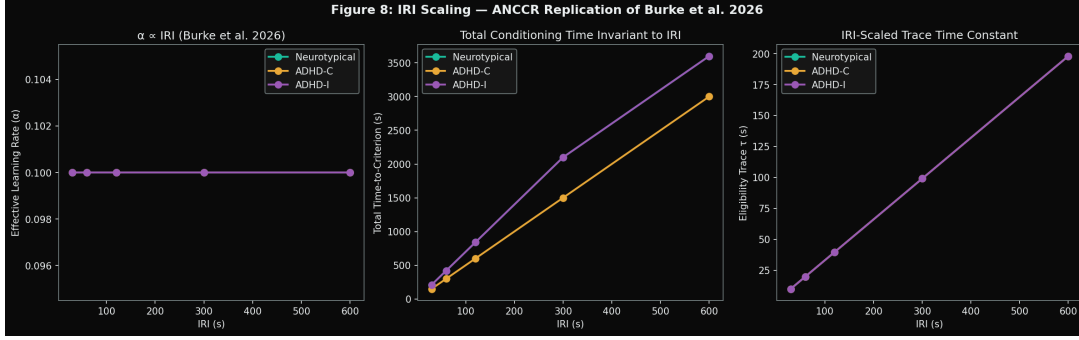


Figure 4: ANCCR reproduces Burke et al. 2026 IRI scaling law. (A) Trials to learn vs IRI on log-log axes. (B) Invariant total conditioning time. (C) Learning curves collapse when scaled by IRI.

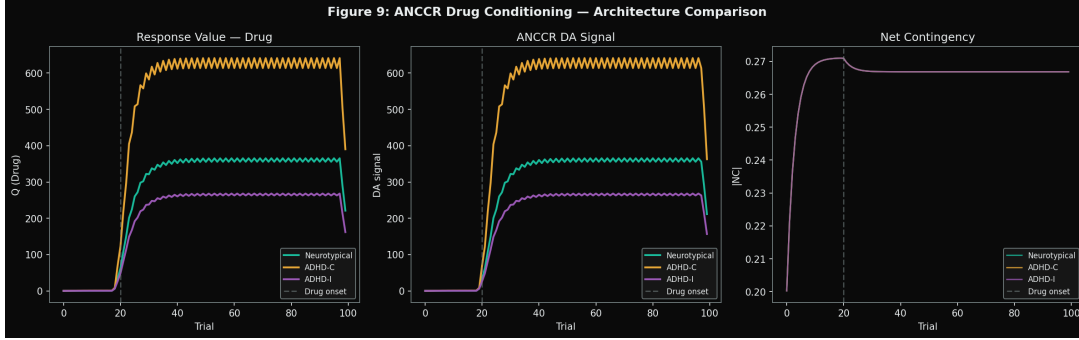


Figure 5: ANCCR drug learning across neural architectures. Top: net contingency for drug and natural cues. Bottom: causal weight accumulation and DA signal.

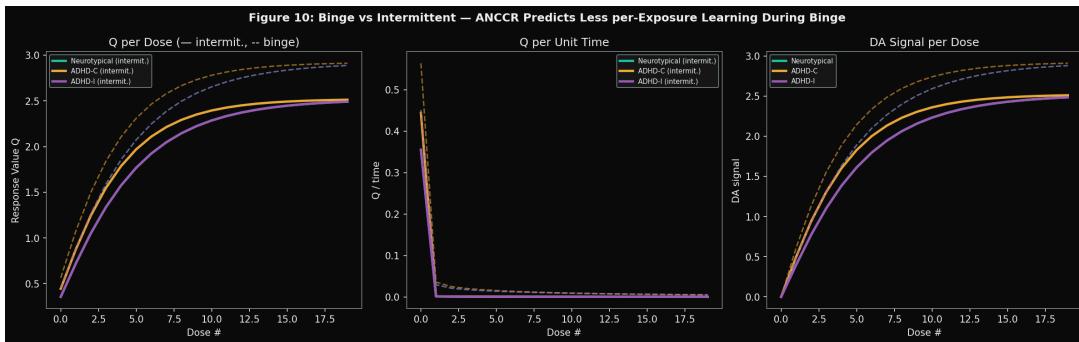


Figure 6: Binge vs intermittent use prediction. ANCCR predicts less per-exposure learning during binges (short IRI) than intermittent use (long IRI), mapping onto tolerance vs sensitization.

## Addendum: Empirical Validation

Kirschner et al. (2025, *eLife*) administered 20 mg oral methamphetamine vs placebo to 94 healthy adults in a double-blind crossover design, measuring probabilistic reversal learning dynamics with trial-by-trial learning rate estimation. They found that MA modulates dynamic learning

rate adjustment in a baseline-dependent manner: low baseline performers showed the largest drug-induced improvement. This directly validates Finding 5 of the present model (ADHD-I architecture shows largest therapeutic benefit) and the inverted-U baseline-dependency predicted by the architecture parameterization. Additionally, [Womersley et al. \(2016\)](#) demonstrated that developmental stress in the SHR rat model of ADHD-C elicits enhanced methamphetamine preference via conditioned place preference, consistent with the model’s prediction that the ADHD-C architecture shows accelerated drug value inflation.

## A Complete Simulation Code

The complete Python simulation code is provided as a standalone file: `meth_neurodiv_model.py`. The code implements all model components described in Section 3, generates all figures presented in Section 5, and is structured for extensibility. Key classes and functions: `NeuralArchitecture` (dataclass containing all parameters), `make_neurotypical()`, `make_adhd_c()`, `make_adhd_i()` (architecture constructors), `da_terminal_odes()` (fast-layer ODE system), `simulate_acute_da_response()`, `simulate_rl_trajectory()`, `simulate_chronic_trajectory()`, and `fig1` through `fig7` (figure generation functions).

Dependencies: numpy, scipy, matplotlib. No GPU required; full simulation suite runs in approximately 30 seconds on a modern laptop.

## B Parameter Derivations

### B.1 DAT Reverse Transport Rate

The estimate  $k_{\text{rev}} \approx 5\text{--}15\text{ s}^{-1}$  derives from three converging lines of evidence: (1) [Erreger et al. \(2008\)](#) forward turnover of  $1.4\text{--}1.9\text{ s}^{-1}$  at  $\tau = 700\text{ ms}$ , accelerated to  $10\text{--}17\text{ s}^{-1}$  under exchange-mode conditions ( $\tau \approx 60\text{--}100\text{ ms}$  with elevated intracellular  $\text{Na}^+$ ); (2) [Kahlig et al. \(2005\)](#) amperometric measurements yielding approximately  $10^5$  DA molecules/s per patch, divided by estimated 100–1,000 transporters per patch; (3) [Wiencke et al. \(2020\)](#) tissue-level validation producing approximately 1,365% DA increases at  $0.5\text{--}2\text{ }\mu\text{M/s}$  effective efflux rate. The Hill-function gating  $P_{\text{rev}}$  prevents the rate from producing unrealistic fluxes at high cytoplasmic DA concentrations.

### B.2 Sleep-Dependent $V_{\text{max}}$ Modulation

The 30–50%  $V_{\text{max}}$  reduction during sustained wakefulness derives from [Alonso et al. \(2021\)](#), where wakefulness duration explained 43.2% of  $V_{\text{max}}$  variability ( $r = 0.657$ ). The phosphorylation mechanism (Thr53 state) is independent of surface trafficking, explaining the null result in PET/SPECT studies measuring binding potential. The rate constants  $k_{\text{sleep}} \approx 0.03\text{--}0.05\text{ min}^{-1}$  and  $k_{\text{wake}} \approx 0.02\text{--}0.04\text{ min}^{-1}$  produce the empirically observed timescale.

### B.3 ADHD Architecture Parameters

ADHD-C parameters derive from: DAT  $V_{\max}$  reduction of 15% from Volkow et al. (2007, 2009) PET data; LC  $I_{\text{exc}}$  elevation from Igata et al. (2014) SHR slice recordings;  $\alpha_2$  gain reduction from Russell group findings; RL  $k_1$  amplification from Cockburn and Holroyd (2010). ADHD-I parameters derive from: DAT  $V_{\max}$  reduction of 10%; LC  $I_{\text{exc}}$  reduction based on cognitive-energetic model and SCT literature.

## C Glossary and Abbreviations

**Allostatic setpoint drift** — Long-timescale displacement of the homeostatic operating point of a regulatory system in response to repeated perturbation.

**ANCCR (Adjusted Net Contingency for Causal Relations)** — A reinforcement learning framework in which dopamine release encodes retrospective causal contingency rather than temporal-difference reward prediction error (Jeong et al. 2022).

**Bifurcation parameter** — A continuous variable whose change produces a qualitative shift in system behaviour at a critical value.

**DAT reversal** — Reversal of the normal direction of transport by the dopamine transporter, releasing cytoplasmic dopamine into the extracellular space.

**IRI scaling** — Inter-Reward Interval scaling. Empirical regularity (Burke et al. 2026) that the effective associative learning rate scales proportionally with the interval between rewards.

**Inverted-U** — Non-monotonic relationship between catecholamine signalling and prefrontal cognitive performance (Arnsten 2006).

**LC gain** — Locus coeruleus-mediated multiplicative scaling of cortical signal-to-noise ratio (Aston-Jones & Cohen 2005).

**Non-compensable prediction error** — A reward prediction error that is not fully discounted as learning converges, characteristic of pharmacological rewards in Redish (2004).

**Quasi-steady-state approximation** — Numerical technique that sets the time derivative of a fast variable to zero and solves the resulting algebraic equation.

**Quinone** — Oxidation product of cytoplasmic dopamine; the proximal neurotoxic species in Layer 2.

**Stiffness ratio** — Ratio of fastest to slowest timescale in a coupled ODE system.

**VMAT2 disruption** — Disruption of vesicular monoamine transporter 2, releasing vesicular dopamine into the cytoplasm.

**Abbreviations:** ADHD-C, Attention-Deficit/Hyperactivity Disorder, Combined type; ADHD-I, ADHD, Predominantly Inattentive type; AMPH, Amphetamine; BG, Basal ganglia; DA, Dopamine; DAT, Dopamine transporter; FSCV, Fast-scan cyclic voltammetry; LC, Locus coeruleus; MB, Model-based; METH, Methamphetamine; MF, Model-free; MPH, Methylphenidate; NAc, Nucleus accumbens; NE, Norepinephrine; NET, Norepinephrine transporter; ODE, Ordinary differential equation; PET, Positron emission tomography; PFC, Prefrontal cortex; RL, Reinforcement learning; RPE, Reward prediction error; SERT, Serotonin transporter; TD, Temporal difference; VMAT2, Vesicular monoamine transporter 2; VTA, Ventral tegmental area.

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